Josue Torres-Fonseca

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EDUCATION

Doctor of Philosophy in Computer Science and Engineering **Expected Graduation: 05/28**
University of Michigan, Ann Arbor, MI Advisor: Dr. Joyce Chai GPA: 4.0

Master of Science in Computer Science and Engineering **Graduated May 2025**
University of Michigan, Ann Arbor, MI Advisor: Dr. Joyce Chai GPA: 4.0

Bachelor of Science in Computer Science, ***Minor: Applied Mathematics*** **Graduated May 2023**
Boise State University, Boise, ID Advisor: Dr. Casey Kennington GPA: 3.889, *Magna Cum Laude*

RESEARCH INTERESTS

Embodied AI safety in multimodal LLMs, VLMs, and VLAs, from high-level planning to low-level control; mechanistic interpretability of multimodal models; language grounding in robots and language models; conversational robots for well-being

RESEARCH EXPERIENCE

PhD Candidate, ***Situated Language and Embodied Dialogue Group*** **Aug 2023 – Present**
University of Michigan, supervised by Dr. Joyce Chai Ann Arbor, MI

- **ReMI: A Conversational Robot to Elicit Resilience Reflection** *HRI 2027, under review*
  - Designed and built an embodied conversational robot that encourages individuals living with cognitive decline to reflect on their resilience, co-designed with 10 dementia experts and 14 people with cognitive impairment
  - Built a **structured dialogue manager** that steps through rapport-building, challenges, and resilience reflection phases; blinded raters scored the system above end-to-end LLM prompting in **35 of 40** simulated conversations
  - **Fine-tuned Qwen3-VL-32B with QLoRA** on 15K dialogue turns **distilled from Gemini 2.5 Pro**, and trained a resilience topic classifier reaching **84% accuracy**; served both on a University of Michigan HPC server with vLLM to protect participant privacy
  - Deployed in **7 lab and 3 in-home studies** with individuals living with cognitive decline; **co-first author**
- **SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal LLMs** *Findings of ACL 2026*
  - Created a **safety-centric dataset** by generating PDDL-planned trajectories that complete household tasks while mitigating six categories of kitchen hazards across 30 AI2-THOR scenes
  - Evaluated **11 state-of-the-art VLMs** on hazard recognition and safe embodied planning, finding models correctly recognized hazards **up to 92%** of the time but fixed them **less than 60%** of the time
  - Developed a **decoupled multi-agent framework** separating hazard recognition from task execution, raising the rate at which Qwen3-VL-32B fixed appliance-misuse hazards **from 0.7% to 71.1%**
  - Open-sourced the code on GitHub and released the dataset on Hugging Face; presented a poster at ACL 2026

Undergraduate Researcher, ***Speech Language and Interactive Machines Group*** **Aug 2021 – Aug 2023**
Boise State University, supervised by Dr. Casey Kennington Boise, ID

- **Symbol and Communicative Grounding through Object Permanence with a Mobile Robot** *SIGdial 2022*
  - Built an **incremental spoken dialogue system** in Python on the retico framework that transcribes and understands speech, detects and grounds objects, and maps them with SLAM on the Anki Cozmo robot
  - Approximated object permanence by extending Cozmo's SLAM map to remember each object it has seen and its location; in an A/B study with 24 participants, this raised the robot's accuracy answering questions about

learned objects **from 70.0% to 82.7%** and **nearly doubled** the questions participants asked

- **HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors** *LREC 2022*
  - Collected and analyzed a dataset of **over 1,000 robot emotional behaviors** described and rated by 32 participants on the Misty II and Cozmo robots
  - Showed a zero-shot language model predicts perceived emotion from the descriptions with **up to 79.3% accuracy**

Undergraduate Researcher, ***Language & Intelligence Group*** **May 2022 – Nov 2022**
Massachusetts Institute of Technology, supervised by Dr. Jacob Andreas Cambridge, MA
MIT Summer Research Program (MSRP)

- Built a training-free variant of MILAN, which describes neurons in vision models in natural language, replacing its 20K human captions with **CLIP** adapted to score WordNet phrases on only the regions where a neuron fires
- Labeled many concrete features in AlexNet and ResNet but struggled with abstract and spatial concepts
- Presented a poster and delivered a lightning talk at the end of the MIT Summer Research Program

Undergraduate Researcher, ***Intelligent Security Group*** **Jun 2020 – Aug 2020**
Texas State University, supervised by Dr. Mina Guirguis San Marcos, TX
NSF Research Experiences for Undergraduates (REU), Smart & Connected Communities

- Designed and built a game and reward system to secure a simulated water tank using game theory
- Implemented check blocks acting as both an intrusion detection and prevention system for the simulated tank

INDUSTRY EXPERIENCE

Software Engineer Intern **May 2021 – Aug 2021**
Paylocity Meridian, ID

- Designed, developed, tested, and deployed new features in project releases as part of a 13-member team, working with product owners and leads to translate use cases into features
- Helped scale tax processes by automating electronic file creation and supporting data exchange with third parties
- Applied Agile and Scrum; gained full-stack experience with JavaScript, React/Redux, C#, .NET, SQL, and Git

TECHNICAL SKILLS

Languages: Python, Java, C, C#, JavaScript, SQL
ML & LLMs: PyTorch, Hugging Face Transformers, PEFT/QLoRA fine-tuning, vLLM, CLIP, HPC GPU clusters
Embodied AI & Robotics: AI2-THOR, PDDL planning, SLAM, retico, Anki Cozmo, Misty II
Software: Git, React/Redux, .NET, Agile/Scrum

PUBLICATIONS

Robot to Elicit Resilience Reflection for Individuals Living with Cognitive Decline **Under Review**
Long-Jing Hsu, **Josue Torres-Fonseca***, Janice Bays, Joyce Chai, and Patricia Alves-Oliveira. 2027. Robot to Elicit Resilience Reflection for Individuals Living with Cognitive Decline. Under review at the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027).* (***Equal contribution, co-first authors**)

SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models **Jul 2026**
***Josue Torres-Fonseca**, Naihao Deng, Yinpei Dai, Shane Storks, Yichi Zhang, Rada Mihalcea, Casey Kennington, Joyce Chai. 2026. SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models. In Findings of the Association for Computational Linguistics: ACL 2026, pages 37190–37211, San Diego, CA, USA. Association for Computational Linguistics.* aclanthology.org/2026.findings-acl.1852

The Mechanistic Emergence of Symbol Grounding in Language Models **Jul 2026**

Shuyu Wu, Ziqiao Ma, Xiaoxi Luo, Yidong Huang, ***Josue Torres-Fonseca****, Freda Shi, Joyce Chai. 2026. The Mechanistic Emergence of Symbol Grounding in Language Models. In Proceedings of the International Conference on Machine Learning (ICML) 2026, Seoul, South Korea.* openreview.net/forum?id=51914Feknq

rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms **Aug 2025**

Anna Manaseryan, Porter Rigby, Brooke Matthews, Catherine Henry, ***Josue Torres-Fonseca****, Ryan Whetten, Enoch Levandovsky, and Casey Kennington. 2025. rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms. In Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue, pages 637–640, Avignon, France. Association for Computational Linguistics.* aclanthology.org/2025.sigdial-1.51

Symbol and Communicative Grounding through Object Permanence with a Mobile Robot **Sep 2022**

Josue Torres-Fonseca*, Catherine Henry, and Casey Kennington. 2022. Symbol and Communicative Grounding through Object Permanence with a Mobile Robot. In Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue, pages 124–134, Edinburgh, UK. Association for Computational Linguistics.* aclanthology.org/2022.sigdial-1.14

HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors **Jun 2022**

Josue Torres-Fonseca *and Casey Kennington. 2022. HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors. In Proceedings of the Thirteenth Language Resources and Evaluation Conference, pages 5739–5748, Marseille, France. European Language Resources Association.* aclanthology.org/2022.lrec-1.617

TALKS & PRESENTATIONS

SafetyALFRED, poster, *ACL 2026*, San Diego, CA **Jul 2026**

Grounding through Object Permanence, talk and poster, *SIGdial 2022*, Edinburgh, UK **Sep 2022**

Interpreting Neural Networks Using Minimal Training Data, poster and lightning talk, *MIT Summer Research Program*, Cambridge, MA **Aug 2022**

HADREB, video presentation, *LREC 2022*, Marseille, France **Jun 2022**

PROFESSIONAL SERVICE

Organizing Committee, *Learning from Situated and Embodied Interaction* Workshop, *COLM 2026*, San Francisco, CA learning-situated-interaction.github.io **Oct 2026**

Reviewer, *ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027)* **2026**

HONORS & AWARDS

Gemini Academic Program Award	**($10,000)**	**Spring 2026 – Spring 2027**
NSF Graduate Research Fellowship	**($159,000)**	**Fall 2023 – Spring 2027**
Boise State University Top Ten Scholar		**Spring 2023**
Barry M. Goldwater Scholarship	**($7,500)**	**Fall 2022 – Spring 2023**
Dean's List, Boise State University		**Fall 2018 – Spring 2023**
Federal Employee Education & Assistance Fund Scholarship	**($1,000)**	**Fall 2018**

MENTORING

Jun Wang, M.S. student, University of Michigan **2026 – Present**

Yuzhang Mei, undergraduate, University of Michigan **2026 – Present**

Avi Bhattacharya, undergraduate, University of Michigan (now Software/ML Engineer, ServiceNow) **2026**

TEACHING EXPERIENCE

Tutor, *Operating Systems*, Boise State University **Jan 2023 – May 2023**

Tutor, *Introduction to Theory of Computation*, Boise State University **Jan 2021 – May 2021**

Tutor, *Data Structures*, Boise State University **Aug 2020 – Dec 2020**

Lab Assistant, *Microprocessors*, Boise State University **Aug 2020 – Dec 2020**

ORGANIZATIONS

Treasurer, *Computer Science and Engineering Graduates*, University of Michigan **Fall 2024 – Spring 2025**

- Managed organization funds and helped organize events

President, *Artificial Intelligence Club*, Boise State University **Aug 2021 – May 2023**

- Ran professional and technical development workshops on Python, machine learning, and data science

President, *Competitive Cybersecurity and Cyber-Physical Club*, Boise State University **Fall 2019 – Spring 2022**

- Competed in the CyberForce Competition at Idaho National Laboratory (46th of 103 teams, Fall 2019)
- Presented on data scraping and reverse engineering with Ghidra; led Capture the Flag activities for members